1 Fundamentals of Microprocessors and Microcomputers

Introduction

A microprocessor is the most important component of a digital computer. It is the Central Processing Unit (CPU) of a digital computer. A digital computer, in which a microprocessor has been provided to act as a CPU, is called a microcomputer. A microcomputer consists of an input device, an output device, a memory and a microprocessor. The microprocessor or CPU of a computer consists of Arithmetic Logic Unit (ALU), accumulator, general and special purpose registers, and timing and control units.

Evolution

During the first two decades, chip complexity increased from Small Scale Integration (SSI) through Medium Scale Integration (MSI) and Large Scale Integration (LSI) to Very Large Scale Integration (VLSI). The block diagram of a microcomputer is illustrated in Fig. 1.1.

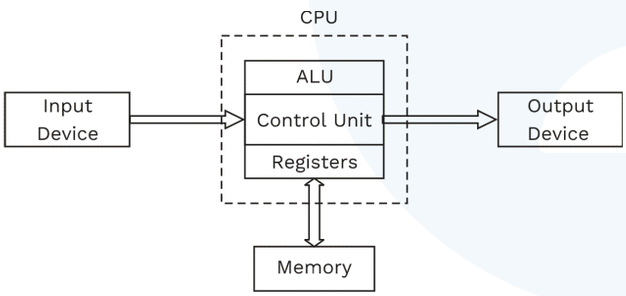


Fig. 1.1 Block Diagram of a Microcomputer

Word Length of Processor

The work length of a processor is the number of data bits the processor can process at a time. For example, an 8085 microprocessor can process only 8 bits.

Evolution of Processors

The first processor, Intel 4004, a 4 bit processor was introduced in 1971 by Intel Corporation, USA. It was followed by 8 bit processors 8008, 8080, and 8085, which were followed by 16 bit processors 8086, 8088, 80186, 80286. Then, Intel came up with 32 bit processors 80386, 486, Pentium, P-II, P-III, and P-IV. Now, 64 bit processors are the latest development in the field of computers. Some 64 bit processors are AMDs, Claw Hammer, Intel's Itanium, etc.

Machine Language

To communicate with a computer, one must provide instruction in binary language, which that is also called machine language.

Assembly Language

In Assembly language, programs are written using English-like words called 'mnemonics' that represent binary instructions of a machine.

Assembler

Assembler is a software program that converts assembly language to machine language.

Manual Assembly

When the program itself acts as an assembler, then this type of assembly is known as manual (or) hard assembly.

Low Level Languages

Machine specific languages are known as low level languages. Machine languages and assembly languages are known as low level languages.

High Level Languages

- Assembly language is specific to a given machine. Programs written in assembly language are not transferred from one machine to another. Therefore, general purpose languages that are machine independent and are called high level languages.
- Examples include BASIC, PASCAL, FORTRAIN, JAVA, C, C^{++}.

Compiler

Compiler reads the entire program first and translates it into the object code that is executed by the processor.

Interpreter

Interpreter reads one instruction at a time, produces its object code, and executes the instruction before reading the next instruction.

Terms Used in Microprocessors

Bit: A digit of the binary number or code is called a bit.

Nibble: The 4 bit binary number or code is called a nibble.

Byte: An 8 bit binary number or code is called a byte.

Word: A 16 bit binary number or code is called a word.

Double word: A group of 32 bit binary numbers or code is called a double word.

Data: The quantity (binary number/code) operated by an instruction of a program is called data. The size of the data is specific as bit, byte, word, etc.

Address: The address is an identification number in binary form for memory locations.

Memory word size: It is the size of binary information that can be stored in a memory location.

Bus: A bus is a group of conducting lines or wires that carries data, address and control signals. Buses can be classified into data bus, address bus, and control bus.

Address bus: A group of conducting lines or wires that carries addresses only is called an address bus.

Data bus: The group of conducting lines that carries only data is called data bus.

Control bus: The group of conducting lines that carry only the control signals is called control buses.

CPU bus: A group of conducting lines that are directly connected to a microprocessor is called a CPU buses. In a CPU bus, the signals are multiplexed, that is, more than one signal is passed through the same line but at different timings.

System bus: A group of conducting lines that carries data, addresses, and control signals in a microcomputer system is called a system bus. Multiplexing is not allowed in a system bus.

Clock: A clock is a square wave generator, which is used to synchronise various devices in the microprocessor and system. Every microprocessor system requires a clock for its functioning. The time taken by the microprocessor and the system to execute an instruction or a program is measured only in terms of time period of its clock.

Practice Exercise

Level I

1. The computer program that converts statements written in high level language to object code is known as:
(A) assembler
(B) operating system
(C) object-oriented software
(D) None of the above

2. Consider the following statements.
I. An assembly language program runs faster than a high level language program to produce the desired result.
II. An assembler that runs on a computer for which it produces object codes is called a resident assembler.

III. A cross-assembler reads the assembly that runs on a computer other than that for which it produces machine codes.
IV. A one-pass assembler reads the assembly language programs only once.

Which of these statements are correct?
(A) I, II, and III
(B) II, III and IV
(C) I and IV
(D) I, II, III, and IV

3. Output of the assembler in machine code is referred to as:
(A) object program
(B) source program
(C) macroinstruction
(D) symbolic addressing

4. Which of the following is correct?
(A) A microprogram is a source program run on microcomputers.
(B) A microprogram is a set of instruction for primitive operations in a system.
(C) A microprogram is a general name for 'macros' used in assembly language programming.
(D) A microprogram is a program of a very small size.

5. What are the sets of commands in a program that are not translated into machine instructions during assembly process called?
(A) Mnemonics
(B) Directives
(C) Identifiers
(D) Operands

6. The commands to the assembler itself are called:
(A) macros
(B) macro-instructions
(C) micro-instructions
(D) pseudo-instructions

Level II

1. Match List-I with List-II and select the correct answer using the codes given below the lists.

List-I

a. Monitor program
b. Assembler
c. Mnemonic
d. Program counter

List-II

I. Used to indicate memory location
II. A combination of letters, symbols, and numerals
III. A program that translates symbolic instructions into the binary equivalent
IV. An operating system

Codes:

	a	b	c	d
(A)	IV	III	II	I
(B)	IV	III	I	II
(C)	III	IV	I	II
(D)	III	IV	II	I

2. **Assertion (A):** Many programmers prefer assembly level programming to machine language programming.
Reason (R): It is possible to efficiently utilise the hardware of the computer in machine language programming.
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

3. Machine instructions are written using which of the following?
(A) Bits 0 and 1 only
(B) Digits 0 and 9 only
(C) Digits 0 and 9, and the capital alphabets A to Z only
(D) Digits 0 to 9, the capital alphabets A to Z and certain special characters

4. Consider the following statements.
Microcontrollers support 'embedded' applications because:
I. the control software is so small that it can be embedded with the microcontroller's ROM.

II. minimum external circuitry is required and the microcontroller can be embedded within a product.

III. a hard disc can be embedded within the computer system supporting the application.

Which of the statements given above is/are correct?

(A) I, II, and III
(B) II and III only
(C) I and II only
(D) III only

5. Match List-I (program required in system software) with List-II (definition) and select the correct answer using the codes given below the lists.

List-I

a. Linker
b. Loader
c. Interpreter
d. Compiler

List-II

I. It is a program that combines smaller programs to form a single program; and also links subroutines with the main program.

II. It is a program that loads machine codes of a program into the system memory

III. It is a program which translates a high-level program into machine code, executes it and reads one statement at a time, and then, goes to the next statements of the program.

IV. It is a program that translates a high-level program into machine language, reads the entire program, and then, executes it

Codes:

	a	b	c	d
(A)	I	I	III	IV
(B)	III	IV	I	II
(C)	I	IV	III	II
(D)	III	II	I	IV

6. A microprocessor program written in assembly language is translated into machine language. The number of instructions in the machine language when compared with the number of instructions in assembly language is:

(A) more
(B) same
(C) less
(D) either more or loss

7. A microcontroller differs from a microprocessor in that it has:

(A) both on-chip memory and on-chip ports
(B) only on-chip memory but not on-chip ports
(C) only on-chip ports but not on-chip memory
(D) neither on-chip memory nor on-chip ports

8. A micro controller differs from a microprocessor in terms of:

(A) I/O interfaces and instruction decoding
(B) memory configuration and I/O interfaces
(C) data bus width and clock speed
(D) memory configuration and instruction decoding

Answer Key

Level I

1. (D) **2.** (D) **3.** (A) **4.** (B) **5.** (B) **6.** (A)

Level II

1. (A) **2.** (B) **3.** (A) **4.** (C) **5.** (A) **6.** (B) **7.** (A) **8.** (B)